



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · 4 categories · Strategic insights from Python, SQL & Power BI

Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Ratings

4

Categories

Demographics, purchase details, behavior signals, and subscription status



Data Preparation in Python

Clean

Imputed missing ratings by category median

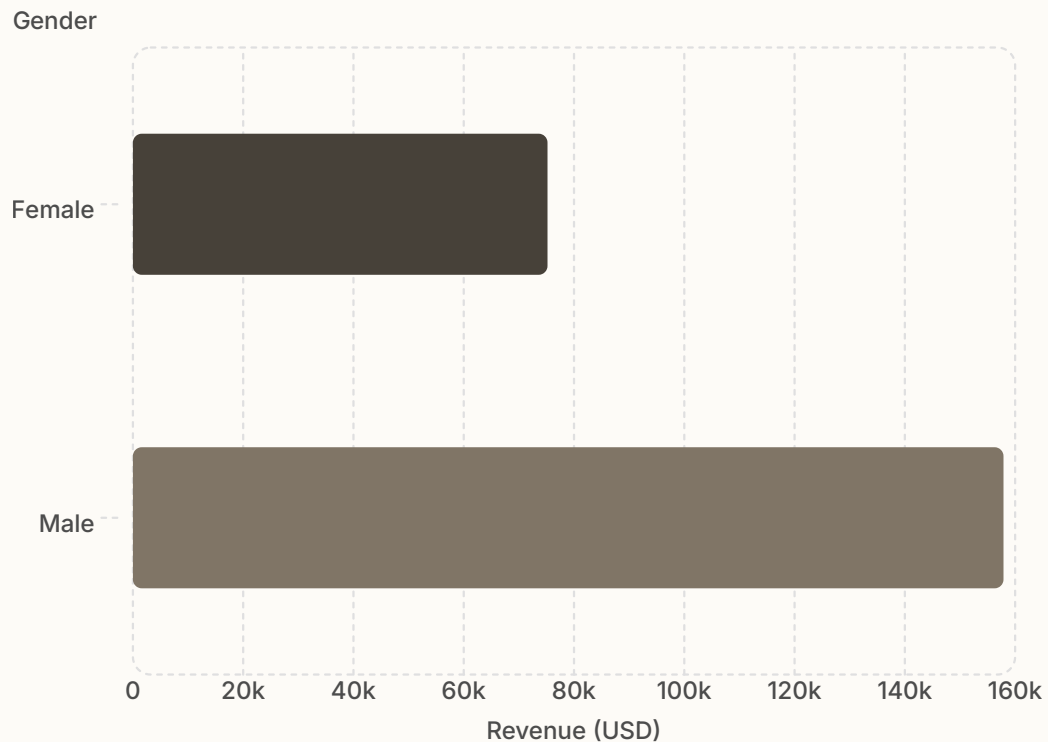
Engineer

Age groups & purchase frequency columns

Integrate

Loaded cleaned data into PostgreSQL

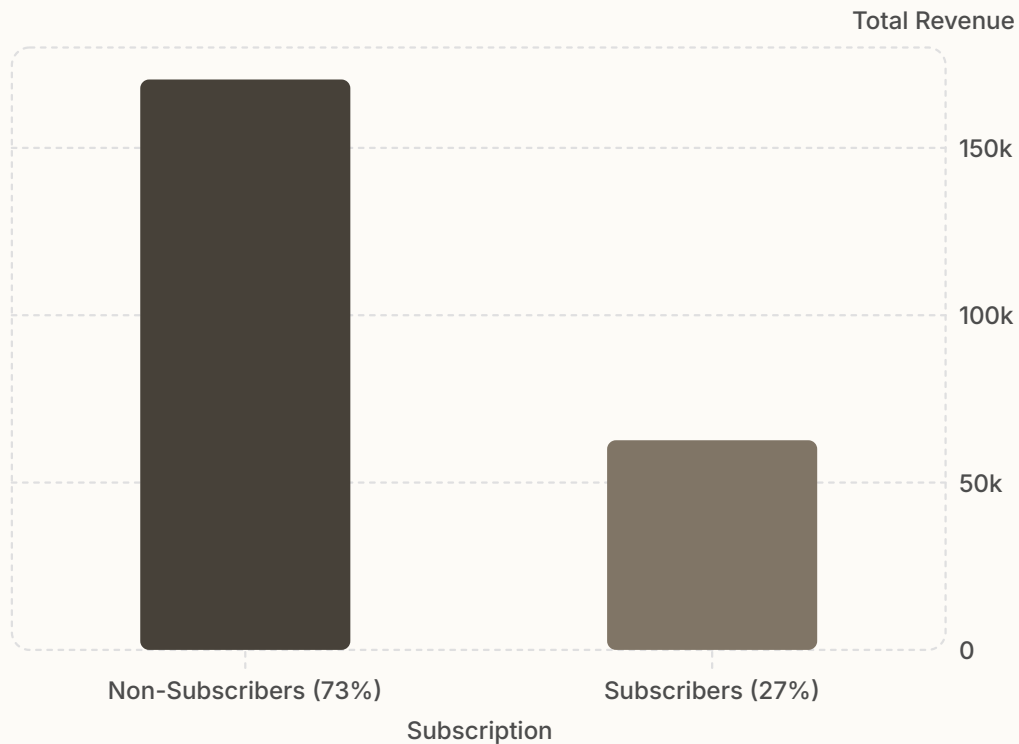
Revenue by Gender



Key Finding

Male customers generated **2× more revenue** than female customers — \$157,890 vs \$75,191

Subscribers vs Non-Subscribers

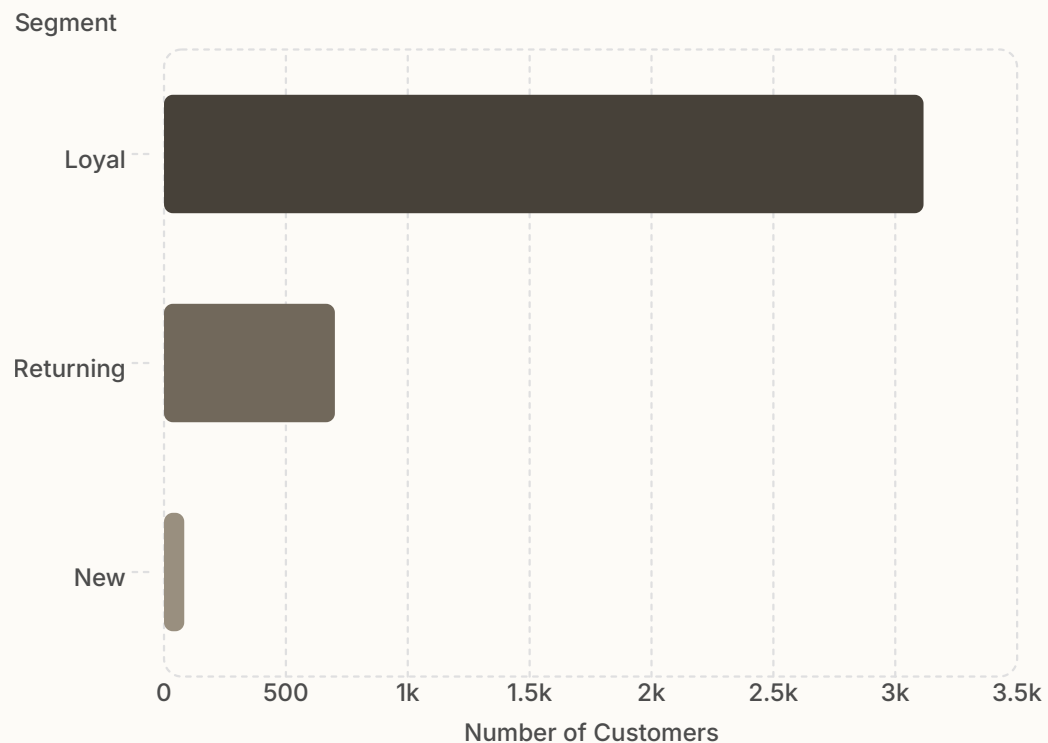


Subscription Gap

Only **27%** of customers subscribe, yet average spend is nearly identical (~\$59)

📌 Opportunity: Convert high-frequency buyers to subscribers

Customer Segmentation



Segment Breakdown

80% Loyal · 18% Returning · 2% New

958 loyal customers have 5+ purchases; 2,518 do not subscribe

Top Products by Category



Accessories

Jewelry · Sunglasses · Belt



Clothing

Blouse · Pants · Shirt



Footwear

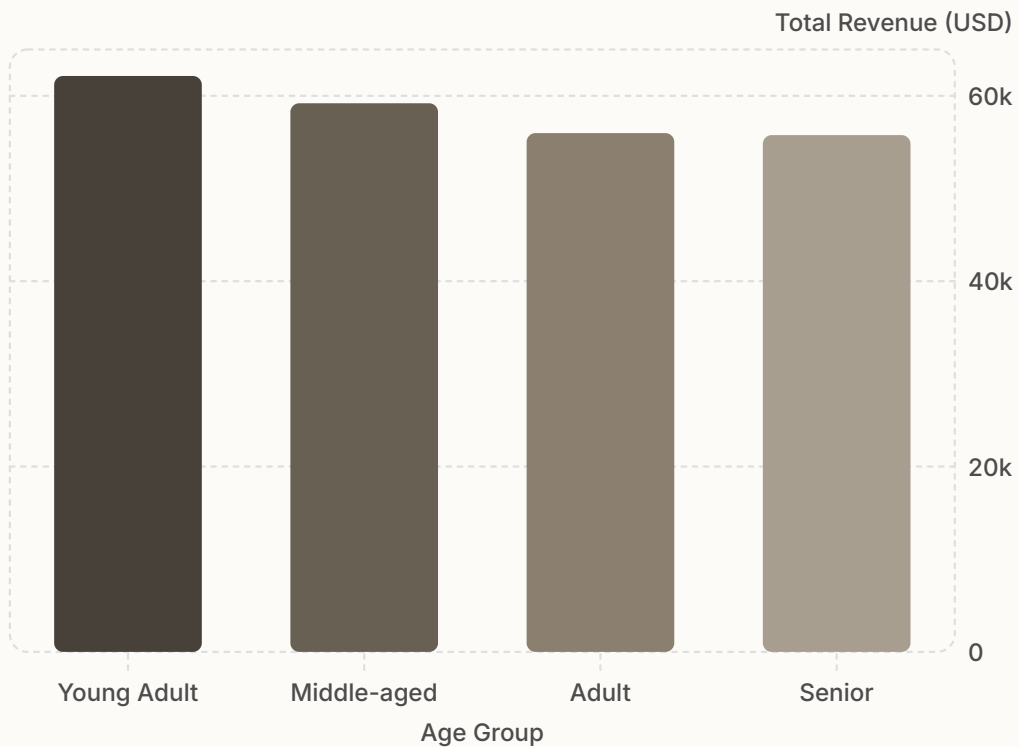
Sandals · Shoes · Sneakers



Outerwear

Jacket · Coat

Revenue by Age Group



Age Insights

Young Adults lead revenue at **\$62,143**. Revenue gap across groups is relatively small (~\$6K spread)



Power BI Dashboard

3.9K Customers

\$59.76 Avg Purchase

3.75 Avg Rating

Clothing ~\$100K Revenue



Business Recommendations

- **Boost Subscriptions**
Promote exclusive subscriber benefits
- **Loyalty Programs**
Reward repeat buyers toward "Loyal" segment
- **Review Discounts**
Balance sales lift with margin control
- **Targeted Marketing**
Focus on high-revenue age groups & express shippers